

BERNHARD PAUS GRÆSDAL

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EDUCATION

- 2022 - Present** | **PHD IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE, MIT**
Student in the Robot Locomotion Group, as part of the MIT Computer Science and Artificial Technology Laboratory (CSAIL).
Research Advisor : Prof. Russ Tedrake
GPA : 4.9 of 5.0
Major : Robotic Manipulation
Minor : Mathematics
- 2016-2021** | **MSC IN CYBERNETICS & ROBOTICS, NTNU**
5 year integrated master's program at the Department of Engineering Cybernetics, NTNU.
GPA : A (of A to F) **Specialization** : Autonomous Systems, Robotics, and Control
- 2019-2020** | **EXCHANGE YEAR, MIT**
Exchange year at the Department of Mechanical Engineering, MIT.
GPA : 5.0 of 5.0 **Subjects** : Robotics, Adaptive Control, Nonlinear Control, Estimation, Machine Learning.

PUBLICATIONS

- 2024** | **"TOWARDS TIGHT CONVEX RELAXATIONS FOR CONTACT-RICH MANIPULATION"**
Bernhard P. Graesdal, Shao Y. C. Chia, Tobia Marcucci, Savva Morozov, Alexandre Amice, Pablo A. Parrilo, Russ Tedrake
Robotics : Science and Systems (RSS)  [Website](#)  [Paper](#)  [Code](#)
- 2025** | **"MIXED DISCRETE AND CONTINUOUS PLANNING USING SHORTEST WALKS IN GRAPHS OF CONVEX SETS"**
Savva Morozov, Tobia Marcucci, Bernhard P. Graesdal, Alexandre Amice, Pablo A. Parrilo, Russ Tedrake
Under review in Robotics : Science and Systems (RSS)
- 2025** | **"PHYSICS-DRIVEN DATA GENERATION FOR CONTACT-RICH MANIPULATION VIA TRAJECTORY OPTIMIZATION"**
Lujie Yang, HJ Suh, Tong Zhao, Bernhard P. Graesdal, Tarik Kelestemur, Jiuguang Wang, Tao Pang, Russ Tedrake
Under review in Robotics : Science and Systems (RSS)  [Preprint](#)
- 2024** | **"GCS* : FORWARD HEURISTIC SEARCH ON IMPLICIT GRAPHS OF CONVEX SETS"**
Shao Y. C. Chia, Rebecca H. Jiang, Bernhard P. Graesdal, Leslie P. Kaelbling, Russ Tedrake
Workshop on the Algorithmic Foundations of Robotics  [Website](#)  [Paper](#)
- 2024** | **"MULTI-QUERY SHORTEST-PATH PROBLEM IN GRAPHS OF CONVEX SETS"**
Savva Morozov, Tobia Marcucci, Alexandre Amice, Bernhard P. Graesdal, Rohan Bosworth, Pablo A. Parrilo, Russ Tedrake
Workshop on the Algorithmic Foundations of Robotics  [Paper](#)
- 2021** | **"FULL NONLINEAR SYSTEM IDENTIFICATION FOR A VERTICAL-TAKEOFF-AND-LANDING UAV"**
Bernhard P. Graesdal
Master's thesis at NTNU (in collaboration with Aviant)  [Thesis](#)  [Thesis model](#)  [Code](#)

WORK EXPERIENCE

2022 SOFTWARE ENGINEER INTERN, COGNITE

6 month internship, spearheading a new project for automatic point cloud segmentation for large, industrial plants.

- > 3D Point Clouds
- > Machine Learning

Python Pytorch Open3D Google Cloud

2019-2021 CTO AND CO-FOUNDER, AVIANT

 MIT News  The Financial Paper (Norwegian)  Today's Business (Norwegian)

- > Team & project management, overseeing the engineering team (5 people)
- > Hardware prototyping and development (Quadcopters & VTOLs)
- > Software development

C++ ROS Gazebo PX4 Python SolidWorks

2019 (Summer) DATA SCIENTIST, AUTODESK FORMA (FORMER SPACEMAKER)

Summer internship. spacemaker.ai

- > Nonlinear optimization of building placements
- > Endpoint development for viewshed algorithm

Python Autodiff Ipopt Flask Javascript

2018-2019 SYSTEM DEVELOPER, CLOUD INSURANCE

Summer internship followed by part-time position. cloudinsurance.io

- > Frontend development with Vue
- > UX design and prototyping

Javascript Typescript Vue HTML CSS

2018 TEACHING ASSISTANT, NTNU

Teaching assistant in linear algebra for two semesters. Met with students twice a week, helping with problem sets and grading of assignments.

OTHER RESEARCH PROJECTS

2020 DYNAMIC SOARING FOR FIXED-WING UAVS

Specialization research project - NTNU [Report](#) [Source](#)

- > Trajectory optimization
- > UAV modeling & energy analysis

Python Drake

2020 COLLISION-FREE MIXED-INTEGER PLANNING FOR UAVS

Independent class project (Underactuated Robotics) - MIT [Video](#) [Source](#) [Report](#)

- > Trajectory optimization
- > Mixed-Integer Semidefinite Programming
- > Time-Varying LQR

C++ Drake Mosek Eigen

2020 ADAPTIVE QUATERNION CONTROL OF QUADROTOR

Independent class project (Nonlinear Control) - MIT [Report](#) [Source](#)

- > Model-Reference Adaptive Control

C++ ROS Gazebo Eigen

GRANTS & SCHOLARSHIPS

- 2021 AKER SCHOLARSHIP**
Norwegian top talent program, similar to NSF Scholarship. akerscholarship.no
- 2019 FULBRIGHT SCHOLARSHIP**
Scholarship for studying at MIT for 2019-2020.
- 2019 JANSON'S LEGAT**
Scholarship for studying at MIT for 2019-2020.
- 2019 NTNU SCHOLARSHIP**
Scholarship for studying at MIT for 2019-2020.

INVITED TALKS & CONFERENCE PRESENTATIONS

- 2024 ROBOTICS : SCIENCE AND SYSTEMS (RSS)**
"Towards Tight Convex Relaxations for Contact-Rich Manipulation"
Conference paper presentation and poster session (in the "Manipulation" session).
- 2022 COGNITE DEVELOPER CONFERENCE**
"Making Point Clouds Smarter"
Invited talk at company conference, presenting the new industrial point cloud segmentation project.

OTHER EXPERIENCE

- 2016 - 2017 ART DIRECTOR**, UKA - NORWAY'S LARGEST CULTURAL FESTIVAL
- 2015 - 2016 OPERATION ASSISTANT AND BORDER GUARD**, THE NORWEGIAN ARMY
- 2014 - 2020 GRAPHIC DESIGNER AND ILLUSTRATOR**, FREELANCE IN SELF-STARTED COMPANY

SKILLS & INTERESTS

-  **Programming** C++ C Python Matlab Golang Javascript Typescript HTML CSS
-  **Frameworks** ROS Gazebo Drake PX4 Git $\LaTeX$ Ubuntu Docker ViM
-  **Hardware** Embedded electronics (CAN, UART, I2C, SPI) Solidworks (basics) 3D-printing
-  **Hobbies** Boulderering & climbing Hiking Literature Music Movies Skiing Running
Illustration Graphic Design

(Compiled on March 17th, 2025).